

From a convincing image to a defensible conclusion

Trustworthy Evaluation Frameworks for AI-Assisted Painting Restoration

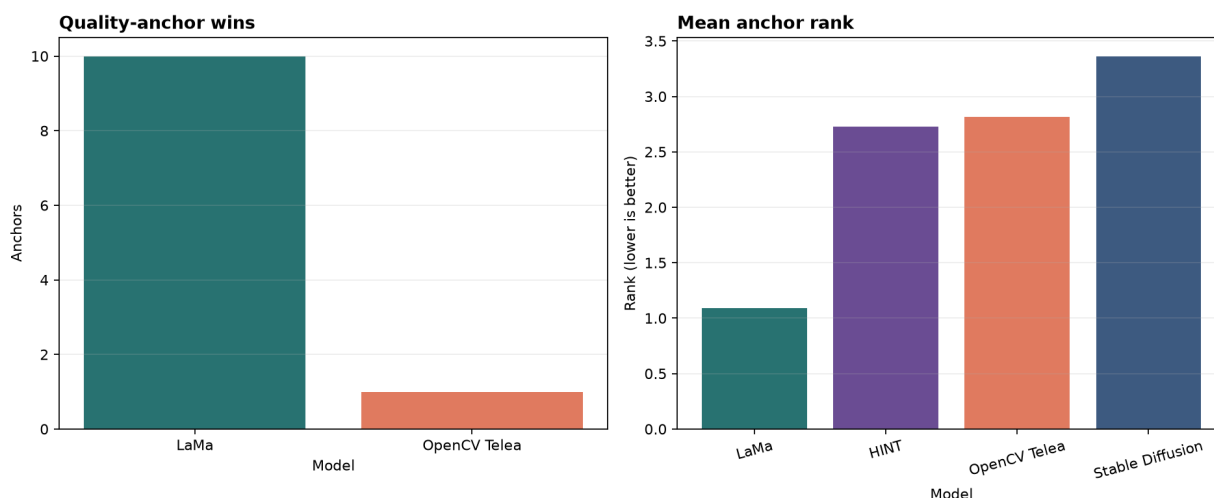
A practical companion to the final eight-room dashboard. Use it to present the experiment, inspect one exact restoration and trace the conclusion to saved evidence.

300 paintings / 3,425 registered cases

2,620 restoration-eligible cases / 13,879 retained report candidates

[Open the Controlled-300 dashboard](#)

Begin in the Exhibition Foyer with **Take the guided tour (about 6 min)**. For a discussion-led meeting, choose **Explore freely** and follow the question that interests your supervisor.



Saved N33 publication figure. LaMa leads 10 of 11 quality anchors; Telea leads crop SSIM. Anchors are separate metric-region comparisons, not a combined score.

Visual plausibility is not restoration trustworthiness.

This 2026-10-02 companion supersedes the pilot-era walkthrough for presentation purposes. It does not modify any frozen notebook, scientific output or N36 package file. Evidence baseline: commit 453b4bfa.

The actual six-minute route

The time labels below are the implemented tour allocations, not measured loading or reading times. Pause wherever a discussion becomes useful.

Room	Guide time	Question
Exhibition Foyer	25 sec	Would you trust a restoration just because it looks convincing?
Study Design	45 sec	What makes a comparison fair?
Metric Framework	55 sec	Can two measurements disagree and both be useful?
Model Gallery	40 sec	What changes when the restoration method changes?
Stability Lab	45 sec	Would the conclusion survive a different seed or condition?
Trustworthiness	45 sec	What should make a human reviewer pause?
Case Explorer	65 sec	Can you connect the image, the numbers, and the caution?
Research Archive	40 sec	Could someone else find the evidence behind your conclusion?

Optional side room: Focused Portrait Review (D02), reached from Trustworthiness. It sits outside the six-minute route and remains a separate study.

For a longer meeting

Suggested discussion route: design and populations; two measurement lenses; the LaMa/Telea contrast; one exact case; then its source record and limitations. Spend longer on the evidence that is challenged rather than trying every filter.

Explore freely

The Foyer offers question-led entry points for looking closely at a painting, comparing methods, examining warning signs or tracing sources. The tour passport is a navigation aid, not an evaluation result.

Build the argument

Exhibition Foyer

Show: Open the tour or free-exploration popup. Introduce the central distinction between a plausible image and a supported restoration conclusion.

Say: The dashboard is a reader of saved evidence, not an inference demo.

Do not claim: A visually attractive output is not evidence of authenticity.

Study Design

Show: Open an experimental path. Explain the 300-painting collection, eligible cases and identity controls before discussing method performance.

Say: Compare like with like and retain the declared population.

Do not claim: Do not count repeated seeds, masks or candidates as independent paintings.

Metric Framework

Show: Keep one restoration fixed; inspect two measurement families and change the region. Open the saved numerical evidence.

Say: Different regions and measurements answer different questions.

Do not claim: Neither feature similarity nor a low pixel error independently establishes trustworthiness.

Model Gallery

Show: Compare method evidence, coverage and the exact values behind a saved plot. Include the Telea crop-SSIM result alongside LaMa's broader lead.

Say: LaMa leads 10 of 11 saved quality anchors in this benchmark.

Do not claim: Do not treat SDXL as fully evaluated or assume zero anchor wins means zero useful cases.

Challenge and trace the argument

Stability Lab

Show: Inspect a repeated-seed comparison or a supported condition change. Check that seed/prompt/experiment scope is consistent.

Say: There are 780 canonical and 245 damage-size four-seed groups.

Do not claim: Disagreement is empirical variability, not calibrated confidence or proof of failure.

Trustworthiness

Show: Open a flag, inspect its rule and supporting evidence, and read unavailable/not-applicable states. Optionally enter D02.

Say: Transparent flags organize human inspection.

Do not claim: A flag is not an expert label. Rendered lightness in D02 is not demographic identity.

Case Explorer

Show: Follow one painting, case, model and candidate. Compare reference, damaged input, mask, restoration and an evidence layer. Then open saved measurements.

Say: Images, numbers and cautions must refer to the exact same candidate.

Do not claim: Missing evidence is not zero, a pass or permission to borrow another candidate's score.

Research Archive

Show: Open the final report or a source record. Inspect a manifest/checksum and the limitations drawers.

Say: Every conclusion should have an accessible source and a stated boundary.

Do not claim: A checksum establishes byte identity, not scientific correctness or a fresh reproduction.

One case, one evidence chain

Use a case that has the required saved evidence. The p018 mixed-damage LaMa example is a useful starting point; the selected identity and available evidence remain authoritative.

Step	Action	Why it matters
1. Fix identity	Read painting, experiment, damage, method and candidate.	Prevents mismatched images or borrowed scores.
2. Establish the task	Compare clean reference, damaged input and mask.	Separates the known target from the missing area.
3. Inspect the result	Compare the restoration and supported evidence layers.	Reveals local changes concealed by a whole-image impression.
4. Read exact rows	Open a measurement: metric, region, damaged/restored values and direction.	A number is meaningful only with its definition and applicability.
5. Test the claim	Inspect a flag, a second measurement or repeated-seed evidence where available.	Keeps disagreement and missingness visible.
6. Trace the record	Open the report/provenance route and find its source.	Connects the demonstration to the saved research record.

A defensible sentence: "For this candidate, this measurement in this region indicates this change. This second diagnostic adds this caution. Neither establishes historical correctness."

If an image or metric is unavailable

Read the availability state and the reason. Do not narrate a blank as a successful result. If remote evidence cannot be loaded during the meeting, use the previously downloaded self-contained report and keep the exact candidate identity visible.

The interface reads saved measurements only; a filter selection does not trigger a new scientific experiment. Cross-room navigation can reload the page; within-room controls are designed for local updates.

Three answers, each with a boundary

RQ1

What additional evidence does a region-aware, multi-metric evaluation framework provide beyond traditional image-similarity metrics when evaluating AI-assisted painting restoration?

Answer: Complementary region-aware metrics expose disagreements; no single or combined score establishes restoration trustworthiness.

Boundary: Metrics and feature similarity are not conservation approval.

RQ2

How do selected inpainting methods differ in restoration quality across controlled artificial damage conditions, and how consistent are these differences across the evaluated paintings?

Answer: LaMa leads 10 of 11 saved quality anchors and Telea leads crop SSIM. HINT is a full deterministic method; SDXL remains bounded. Conclusions apply to this controlled benchmark.

Boundary: No universal model superiority or independently established art-historical style effect.

RQ3

How can repeated-candidate disagreement be used to characterize the stability of stochastic painting restorations, and how does it relate to other restoration-quality diagnostics?

Answer: The 780 canonical and 245 damage-size groups characterize repeated-seed variation. Disagreement supports inspection, not calibrated confidence or historical correctness.

Boundary: Only supported repeated-seed groups; deterministic methods do not acquire seed uncertainty.

Exact RO evidence and source-row identities

Do not infer a universal effect from a particular condition, or a causal relationship from diagnostic association. The finalized study supports evidence-guided evaluation, not automated conservation approval.

Traceability, limitations and review questions

What is complete

The numbered N01-N36 pipeline, including N12A, is complete. D01 remains separate method-selection evidence and D02 a supplemental portrait audit. N36 records 859 passes, 16 warning nonpasses and zero blockers.

What remains qualified

The 16 nonpasses preserve four N35 dependency warnings, ten unmeasured owner-accepted qualifications, the historical N29 manifest discrepancy and the N35 notebook-source mismatch. No exhaustive browser, performance-budget, clean-Linux reproduction or independently attested server-revision claim is made.

Use these sources when challenged

[N33 final evaluation report](#) - scientific narrative and embedded figures.

[N36 supervisor synthesis](#) - claims, populations and interpretation limits.

[Reproducibility appendix](#) and [limitations record](#) - source identities and disclosed exceptions.

HTML links on GitHub show source. Download and open the HTML locally, or open it through the dashboard. Distribute all 134 files in the N36 output directory, not the package subfolder alone.

Ask the supervisor:

Are the three bounded answers convincing?

Which figures and cases best support the thesis narrative?

How prominently should SDXL feasibility and D01/D02 be presented?

Are the limitations and future-work boundaries sufficiently explicit?

The implementation is complete. Thesis discussion, presentation choices and any future publication remain human decisions.